

YIHANG TAO

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EDUCATION

City University of Hong Kong

PhD, Department of Computer Science
Supervisor: Prof. Yuguang "Michael" Fang
JC STEM Lab of Smart City, WINET Laboratory

Hong Kong SAR
Sep. 2024 - Now

Shanghai Jiao Tong University

Master, School of Computer Science
Outstanding Graduate (2024)

Shanghai, China
Sep. 2021 - Apr. 2024

Southeast University

Bachelor, School of Information Engineering
GPA: 3.89/4.0

Nanjing, China
Sep. 2017 - Jul. 2021

RESEARCH INTERESTS

Autonomous Driving, Spatial Intelligence, Generative Model, AI Security.

PUBLICATIONS

First Author (* Equal contribution)

[**Under Review**] **Y. Tao**, Y. Guo, Z. Fang, H. An, and Y. Fang. "FRUC: Feedforward Dynamic Scene Reconstruction from Uncalibrated Collaborative Driving Views," *Under Review*.

[**Under Review**] **Y. Tao***, Y. Guo*, S. Hu, Y. Ma, Z. Fang, S. Kwong, and Y. Fang. "V2XCrafter: Learning to Generate Driving Scene Across Agents," *Under Review*.

[**CVPR'26**] **Y. Tao**, S. Hu, H. An, Z. Fang, H. Cao, and Y. Fang. "Learning Mutual View Information Graph for Adaptive Adversarial Collaborative Perception," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [**CCF-A**]

[**TDSC**] **Y. Tao***, S. Hu*, Y. Hu, H. An, H. Cao, and Y. Fang. "GCP: Guarded Collaborative Perception with Spatial-Temporal Aware Malicious Agent Detection," *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. [**CCF-A**]

[**ICRA'25**] **Y. Tao**, S. Hu, Z. Fang, and Y. Fang. "Directed-CP: Directed Collaborative Perception for Connected and Autonomous Vehicles via Proactive Attention," *IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, USA, 2025. [**CCF-B**]

[**AAAI'25**] S. Hu*, **Y. Tao***, G. Xu, Y. Deng, X. Chen, Y. Fang, and S. Kwong. "CP-Guard: Malicious Agent Detection and Defense in Collaborative Bird's Eye View Perception," *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI)*, Philadelphia, USA, 2025. (**Oral Presentation**, <5%) [**CCF-A**]

[**TGCN**] **Y. Tao**, J. Wu, Q. Pan, A. K. Bashir, and M. Omar. "O-RAN-Based Digital Twin Function Virtualization for Sustainable IoV Service Response: An Asynchronous Hierarchical Reinforcement Learning Approach," *IEEE Transactions on Green Communications and Networking (TGCN)*, vol. 8, no. 3, pp. 1049-1060, Sep. 2024.

[**GLOBECOM'23**] **Y. Tao**, J. Wu, X. Lin, S. Mumtaz, and S. Cherkaoui. "Digital Twin and DRL-Driven Semantic Dissemination for 6G Autonomous Driving Service," *IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, Dec. 2023, pp. 2075-2080. [**CCF-C**]

[**LNET**] **Y. Tao**, J. Wu, X. Lin, and W. Yang. “DRL-Driven Digital Twin Function Virtualization for Adaptive Service Response in 6G Networks,” *IEEE Networking Letters (LNET)*, vol. 5, no. 2, pp. 125-129, Jun. 2023.

Collaborative Author

[**Under Review**] Z. Fang*, Y. Guo*, F. Liu, Y. Zhang, **Y. Tao**, S. Hu, W. Ding, and Y. Fang. “Agent-Centric Visual Reinforcement Learning under Dynamic Perturbations,” *Under Review*.

[**CVPR’26**] H. An, Y. Xiaohui, G. Hua, **Y. Tao**, H. Cao, X. Yu, and Y. Fang. “RecoverMark: Robust Watermarking for Localization and Recovery of Manipulated Faces,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [**CCF-A**]

[**TDSC**] H. An, G. Hua, W. Du, H. Cao, **Y. Tao**, G. Xu, S. Rahardja, and Y. Fang. “Decoder Gradient Shields: A Family of Provable and High-Fidelity Methods Against Gradient-Based Box-Free Watermark Removal,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. [**CCF-A**]

[**TMC**] S. Hu, **Y. Tao**, G. Xu, X. Qian, Y. Deng, X. Chen, S. Kwong, and Y. Fang. “CP-uniGuard: A Unified, Probability-Agnostic, and Adaptive Framework for Malicious Agent Detection and Defense in Multi-Agent Embodied Perception Systems,” *IEEE Transactions on Mobile Computing (TMC)*, 2025. [**CCF-A**]

[**NeurIPS’25**] Y. Guo, S. He, Y. Lu, H. An, **Y. Tao**, H. Zhu, J. Liu, and Y. Fang. “Neptune-X: Active X-to-Maritime Generation for Universal Maritime Object Detection,” *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, **Spotlight**, San Diego, USA, Dec. 2025. [**CCF-A**]

[**NeurIPS’25**] S. Hu, X. Han, J. Jiang, **Y. Tao**, Z. Fang, Y. Dai, S. Kwong, and Y. Fang. “Distribution-Aligned Decoding for Efficient LLM Task Adaptation,” *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, San Diego, USA, Dec. 2025. [**CCF-A**]

[**GLOBECOM’25**] S. Hu, Y. Ma, **Y. Tao**, Z. Fang, Z. Fang, Y. Deng, S. Kwong, and Y. Fang. “Task-Aware Parameter-Efficient Fine-Tuning of Large Pre-Trained Models at the Edge,” *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, Dec. 2025. [**CCF-C**]

[**JSAC**] Z. Fang, J. Wang, Y. Ma, **Y. Tao**, Y. Deng, X. Chen, and Y. Fang. “R-ACP: Real-Time Adaptive Collaborative Perception Leveraging Robust Task-Oriented Communications,” *IEEE Journal on Selected Areas in Communications (JSAC)*, 2025. [**CCF-A**]

[**TDSC, Major Revision**] S. Hu, **Y. Tao**, Z. Fang, G. Xu, Y. Deng, S. Kwong, and Y. Fang. “CP-Guard+: A New Paradigm for Malicious Agent Detection and Defense in Collaborative Perception,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, *Major Revision*. [**CCF-A**]

[**TMC**] Z. Fang, Z. Lin, S. Hu, **Y. Tao**, Y. Deng, X. Chen, and Y. Fang. “Dynamic Uncertainty-aware Multimodal Fusion for Outdoor Health Monitoring,” *IEEE Transactions on Mobile Computing (TMC)*, 2026. [**CCF-A**]

HONORS & AWARDS

Outstanding Academic Performance Award (OAPA), City University of Hong Kong	Aug. 2025
IEEE Robotics and Automation Society (RAS) Travel Grant for ICRA’25	Mar. 2025
Outstanding Graduate, Shanghai Jiao Tong University	May 2024
WEICHAi POWER Scholarship, Shanghai Jiao Tong University	Oct. 2023
Excellent League Member, Shanghai Jiao Tong University	Apr. 2022
National Student Research Training Program Excellence Award (Leader)	Oct. 2020

Excellence Prize, 2nd International Data Competition, IKCEST (**top 3%**) Oct. 2020
 Sun Qingyun Innovation Scholarship, Southeast University (<**1% annually**) Jun. 2020
 Finalist, 36th Mathematical Contest in Modeling (MCM), COMAP (**top 1%**) Apr. 2020
 First Prize, 12th National Information Security Contest, China (**top 8%**) Aug. 2019
 First Prize, 15th Advanced Mathematics Competition, Jiangsu (**top 10%**) Aug. 2018

ACADEMIC SERVICE

Program Committee Member:

CVPR, ECCV, ICML, NeurIPS, ICLR, ACM MM, IROS, ICASSP, IJCNN

Journal Reviewer:

IEEE TMC, IEEE TITS, IEEE TCE, Pattern Recognition, Neural Networks, EAAI, IEEE JBHI, IEEE LNET

Journal Administrator:

IEEE TMLCN

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C++, C#, JavaScript, Verilog HDL, $\text{\LaTeX}$, Markdown

Deep Learning Frameworks: PyTorch, TensorFlow, OpenMMLab (MMDetection, MMSegmentation)

Tools & Platforms: Git, Docker, Linux, CARLA Simulator, SUMO, ROS

Languages: English (IELTS 7.0), Chinese (Native)